Algorithm Design Techniques: Dynamic Programming-1

All Pairs Shorted Paths

Floyd-Warshall Algorithm

Transitive Closure

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Reminder: Algorithm Design Methods

- Brute Force
- Greedy
- Divide and Conquer
- Dynamic Programming (DP)
- Backtracking
- ……

- Dynamic Programming is a powerful technique that can be used to solve many optimization problems in polynomial time for which a naive approach would take exponential time.

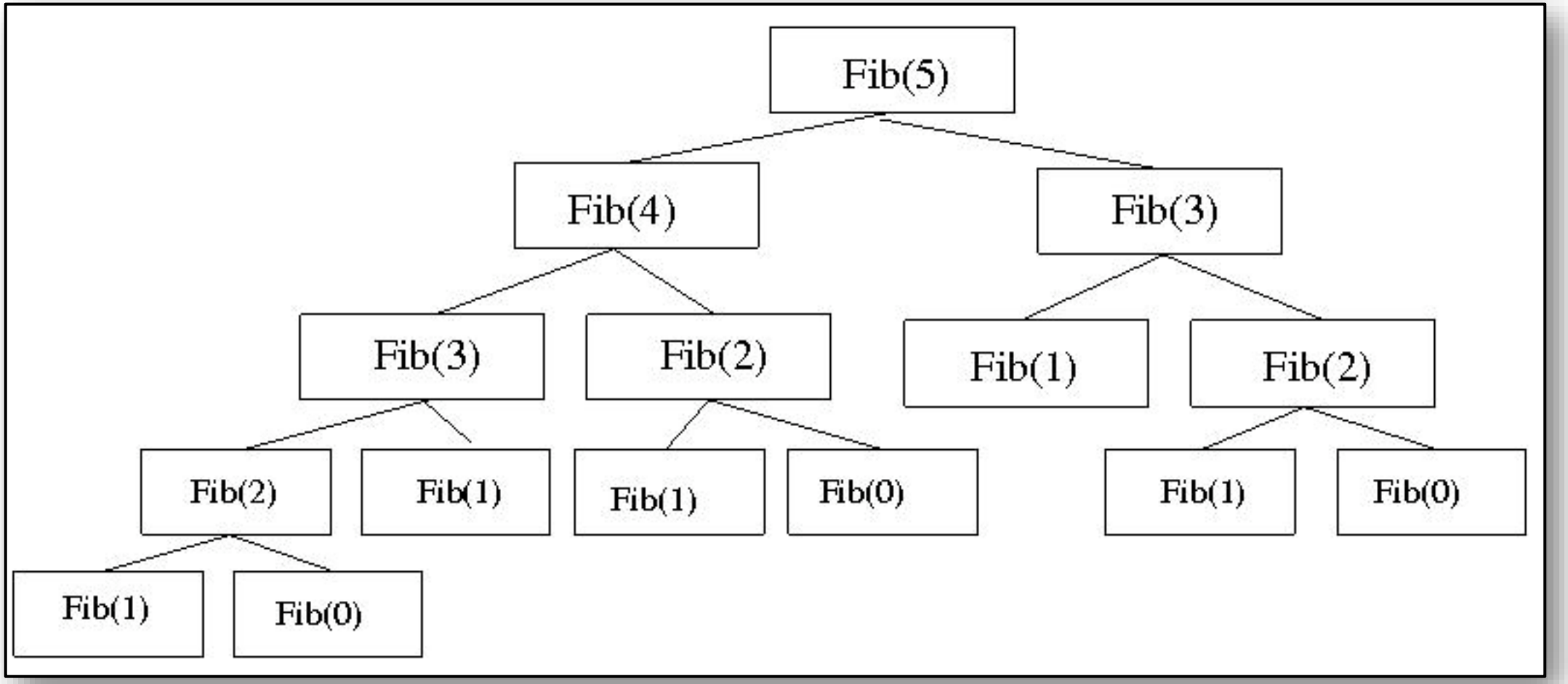
Warning : Dynamic programming → planning over time , not computer programming

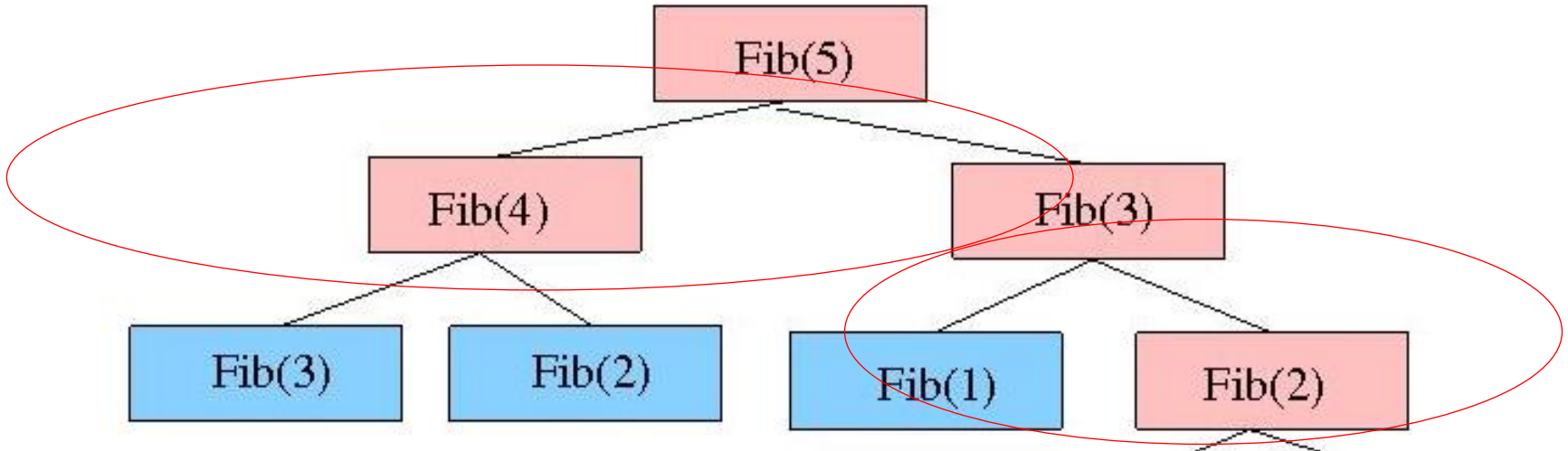
Introduction to Dynamic Programming

- We have looked at several algorithms that involve recursion.
- In some situations, these algorithms solve fairly difficult problems efficiently
  - *BUT* in other cases, they are inefficient because they **recalculate certain function values** many times.
- One example of this is the Recursive Fibonacci algorithm.
- Remember the problem of recursive Fibonacci:
  - ***Lots and lots* of calls to Fib(i-1) and Fib(i-2) are made for i>2**
- How to find for example Fib(10) more efficiently ?

Reminder : Fibonbacci Tree

What's the problem?





Already calculated ! Several redundant computations.

Dynamic Programming : Efficient Fibonacci Function

- First make a chart (Table) and store the values :
  - $F_1 = 1$, $F_2 = 1$, $F_3 = 2$, $F_4 = 3$, $F_5 = 5$, $F_6 = 8$, $F_7 = 13$, $F_8 = 21$, $F_9 = 34$, $F_{10} = 55$…
  - Calculate F_3 by adding F_1 and F_2, then F_4, by adding F_3 and F_4, etc.
- This way, we can calculate any Fibonacci number by simply adding stored values.
- The basic idea of dynamic programming is to avoid making redundant method calls.
  - Instead, **store the answers** to all necessary method calls in memory and simply look these up whenever needed.
  - This is called **Memoization** (forming a memo)

  → Store the solution of these sub-problems so that we do not have to solve them over and over again

Fibonacci with Memoization

```
int memo[] = empty
int fib(n) {
  if(memo[n] != empty)  return memo[n];  // should be empty in the beginning
  if(n==1 || n==2) return 1;
  memo[n] = fib(n-1)+fib(n-2);
  return memo[n];
}
```

Fibonacci: Dynamic Programming Solution

```
//Input integer n,output nth fib.number
//Previous values are stored in the array fib[]
fibon(int n)
{
   int fib[n+1];    //Declare the array
   fib[0] = 0; fib[1] = 1; //initial values are set
   for (i=2; i<n; i++) fib[i] = fib[i - 1] + fib[i - 2];
   return fib[n];
}
```

<u>Complexities:</u>

Time Complexity: O(n) , Space Complexity : O(n)

Note that the complexity is just linear, not exponential.

```
//Fibonacci Series using Dynamic Programming
#include<iostream>
using namespace std;
int fib(int n)
{  // Declare an array to store Fibonacci numbers.
  int f[n+1];   int i;
   // 0 th and 1st number of the series are 0 and 1
  f[0] = 0;   f[1] = 1;
  for (i = 2; i <= n; i++)
     // Add the previous 2 numbers in the series and store it
      f[i] = f[i-1] + f[i-2];
  return f[n];
}
 int main ()
{
  int n = 18;
  cout<<"n =  "<< n<<"   "<<"fib n  = "<<fib(n)<<endl;
  return 0;
}
//n =  18   fib n  = 2584
```

Dynamic Programming

- DP is essentially an algorithm design technique for **optimization problems**: Often minimizing or maximizing.
- Like divide and conquer, DP solves problems by combining solutions to subproblems.
- **However, in DP, subproblems are not independent.**
  - Subproblems may share subsubproblems,
  - However, solution to one subproblem may not affect the solutions to other subproblems of the same problem.

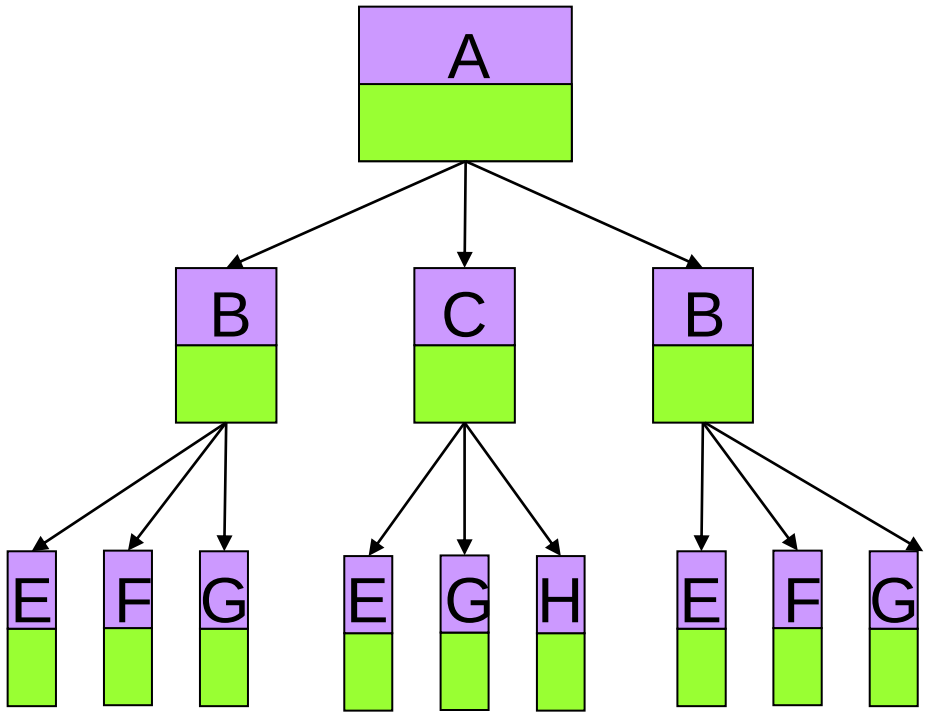
Dynamic Programming : How it Works ?

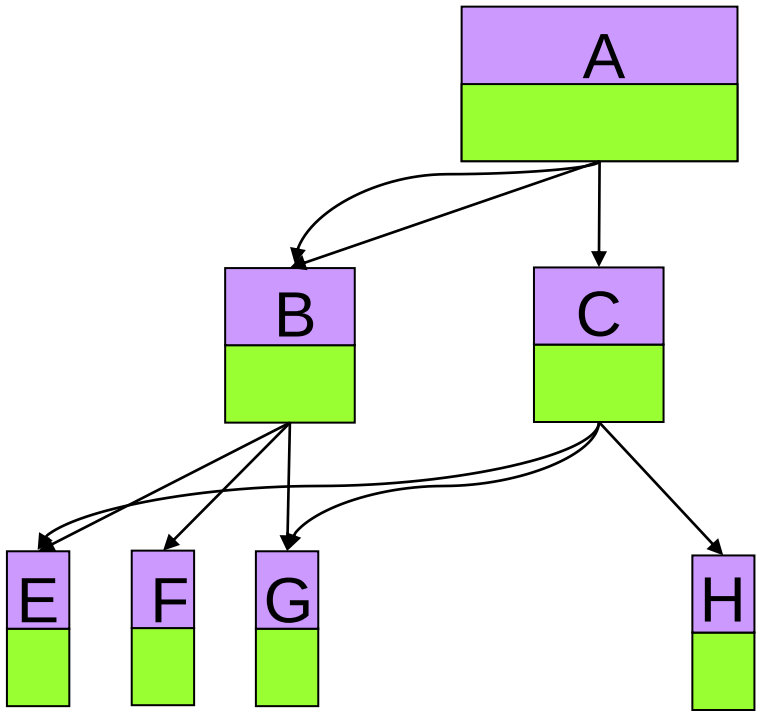
- **Memoization**: Store the solution of the subproblems so that we do not have to solve them again.
- Dynamic programming and memoization work together. Most of the problems are solved with two components of dynamic programming :
  - Recursion – Solve the subproblems recursively
  - Memoization – Store the solution of these subproblems so that we do not have to solve them again

→ DP reduces computations bottom up or top down by

- **Solving** subproblems
- **Storing** solution to a subproblem the first time it is solved.
- **Looking** up the solution when subproblem is encountered again.

Dynamic Programming:





Actual computations with memorization: No redundant repetitions.

Properties of Dynamic Programming

- Suppose we have a problem that we can break it to **smaller subproblems** having the following properties:
- **Optimal Substructure**
  - An optimal solution to a problem contains within it an optimal solution to subproblems.
  - Optimal solution to the entire problem is built bottom-up from optimal solutions to subproblems
- **Overlapping Subproblems**
  - If a recursive algorithm revisits the same subproblems over and over $\Rightarrow$ the problem has overlapping subproblems
- If the problem has these two properties, then we can attempt to solve that problem using DP.

Dynamic Programming vs Brute Force

Dynamic programming is considered to be a "Careful brute force"

Example:

Dynamic Programming : You want 5 things. You search for them in 100 shops. You save names and locations of the shops and the items you can get from there. If you need those 5 items , you don't search every time in 100 shops. You check the saved locations and go only to those specific shops

Brute Force : You search in 100 shops every time.

Dynamic programming vs Divide & Conquer

- Comparing to divide-and-conquer:
  - Both partition the problem into sub-problems
  - Divide-and-conquer partition problems into **independent sub-problems**.
  - Dynamic programming is applicable only when the **sub-problems are dependent.**

Dynamic programming vs Greedy

- Comparing to greedy method:
  - Both are applicable when a problem exhibits optimal substructure
  - In greedy algorithms, we use optimal substructure in a **top-down fashion**, but **without finding optimal solutions to sub-problems.**
  - We first make a choice that looks best at the time-and then solve a resulting sub-problem.
  - In dynamic programming, we use **optimal substructure in a bottom-up fashion:**
    - We first find optimal solutions to sub-problems and, having solved the sub-problems, we find an optimal solution to the problem.

DP Applications: Shortest Path Problems

- Optimal Substructure (Principle of optimality): Suppose that in solving a problem, we have to make a sequence of decisions D_1, D_2, …, D_n. If this sequence is optimal, then the last k decisions, $1 < k < n$ must also be optimal.
- Example: The shortest path problem

  If i, i_1, i_2, …, j is a shortest path from i to j, then i_1, i_2, …, j must be a shortest path from i_1 to j

→If a problem can be described by a weighted graph, then shortest path problem can be solved by dynamic programming.

All Pairs Shortest Paths

- Given **a weighted graph**, we want to know **the shortest path from any vertex in the graph to all others.**

  - The **Floyd-Warshall algorithm** determines the shortest paths between all pairs of vertices in a graph.

All Pairs Shortest Paths:

- Aims to compute the shortest path from each vertex to every other vertex.
- It Uses Dynamic Programming methodology to solve the problem.

All Pairs Shortest Paths : Floyd-Warshall Algorithm

- *The problem:* find the shortest path between every pair of vertices of a graph

- *The graph* may contain negative edges but no negative cycles(Cycles whose edges sum to a negative value)
  Adjacency matrix representation: A weight matrix W:

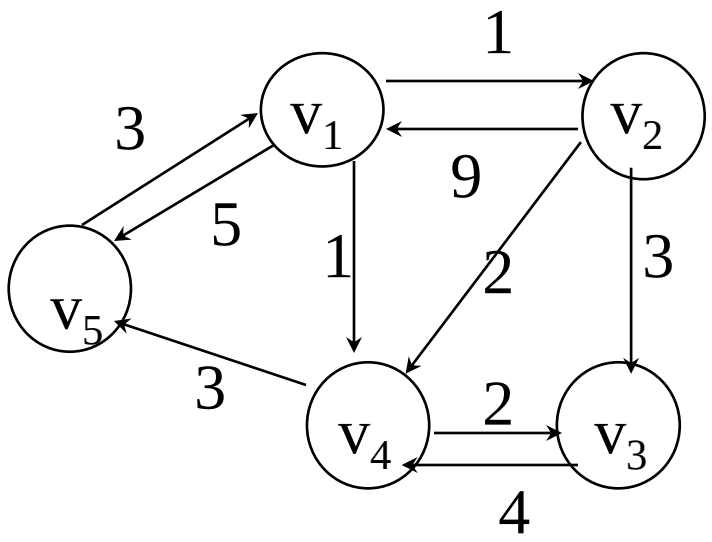
  W(i,j)= weight of edge (i,j)
  W(i,j)=0 if i=j.
  W(i,j)=∞ if there is no edge between i and j.

Example :Weight Matrix Representation

$w_{i,j}$: Distance or cost of (i,j)

W=

	1	2	3	4	5
1	**0**	**1**	∞	**1**	**5**
2	**9**	**0**	**3**	**2**	∞
3	∞	∞	**0**	**4**	∞
4	∞	∞	**2**	**0**	**3**
5	**3**	∞	∞	∞	**0**



The subproblems

- Let $D^{(k)}[i,j]$=The weight of a shortest path from v_i to v_j that uses only vertices from $\{v_1,v_2,\ldots,v_k\}$ as intermediate vertices in the path.

  $D^{(0)}$= *W, initial distances*

  $D^{(k)}$ = Distances at *Intermediary step k*

  $D^{(n)}$= *T*he goal matrix. Final shortest distances

- How do we compute $D^{(k)}$ from $D^{(k-1)}$ ….?

→We need to compute : $D^{(0)}$, $D^{(1)}$,……, $D^{(n)}$

The Recursive Definition

There are two possibilities:

#1 ➡ $D^{(k)}[i,j] = D^{(k-1)}[i,j]$ (k has no effect, not part of the min. path)

#2 ➡ $D^{(k)}[i,j] = D^{(k-1)}[i,k] + D^{(k-1)}[k,j]$ (k is part of the min.path)

Therefore, we conclude that the shortest path must be smaller of 1 or 2:

$$D^{(k)}[i,j] = \min\{ (D^{(k-1)}[i,j]) , (D^{(k-1)}[i,k] + D^{(k-1)}[k,j]) \}$$

Floyd-Warshall algorithm :Explanation

Let $d_{ij}^{(k)}$ be the weight of a shortest path from vertex i to vertex j with all intermediate vertices in the set {1,2,…,k}. A recursive definition is given by

$$d_{ij}^{(k)} = \begin{cases} w_{ij} & \text{if } k=0, \\ \min(d_{ij}^{(k-1)}, d_{ik}^{(k-1)}+d_{kj}^{(k-1)}) & \text{if } k \geq 1. \end{cases}$$

Floyd-Warshall Algorithm:Explanation

For all k in 0…n ,we have:

$d_{ij}^{(0)} = w_{ij}$ **if k=0, means that there are no intermediate vertices at all**

$d_{ij}^{(k)} = \min(d_{ij}^{(k-1)}, d_{ik}^{(k-1)} + d_{kj}^{(k-1)})$ **if $k \geq 1$**

Result: $D^{(n)} = (d_{ij}^{(n)})$

→The matrix $D^{(n)}=(d_{ij}^{(n)})$ gives the final answer:

$d_{ij}^{(n)}$= Shortest for all i,j pairs in V .

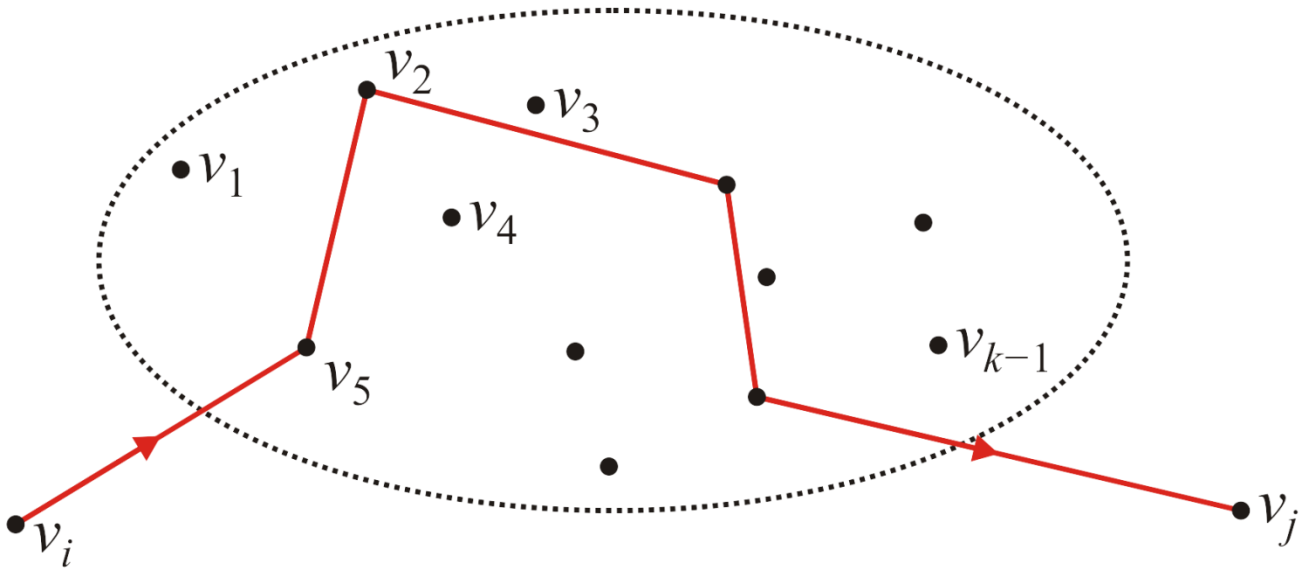
Because all intermediate vertices are in the set {1,2,…,n}.

The General Step

Define $d_{i,j}^{(k-1)}$ as the shortest distance, but only allowing intermediate visits to vertices in the set: $v_1, v_2, \ldots, v_{k-1}$

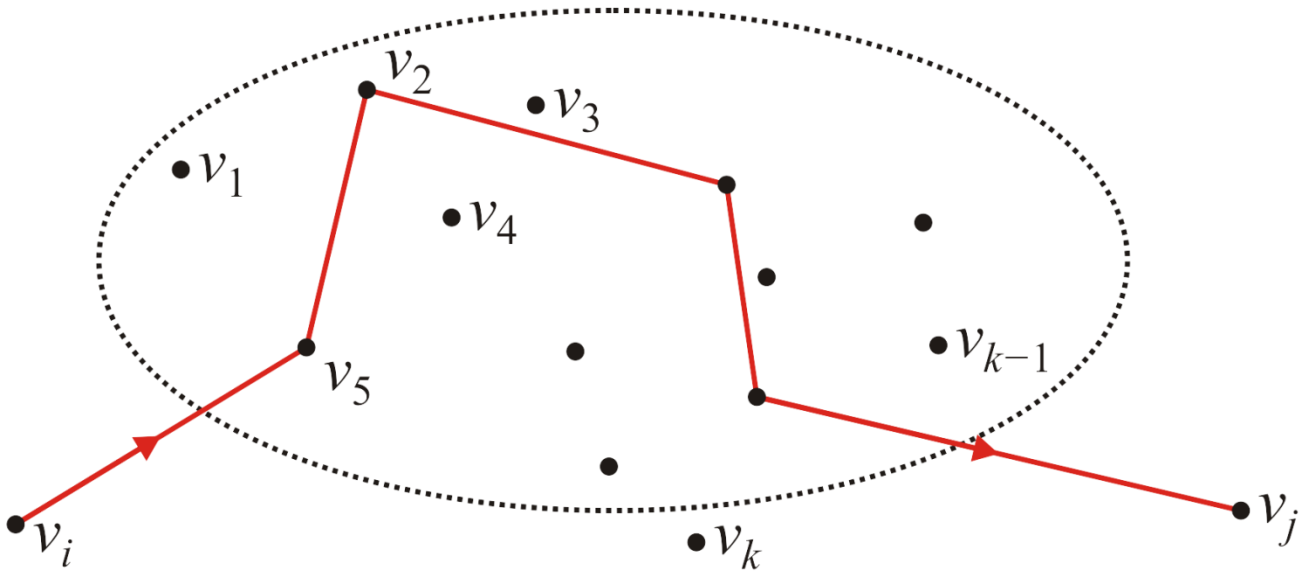
- Suppose we have an algorithm that has found these values for all pairs

Can we do better by including v_k now?



The General Step

How could we find $d_{i,j}^{(k)}$; that is, the shortest path allowing intermediate visits to vertices v_1, v_2, …, v_{k-1} and v_k?



The General Step

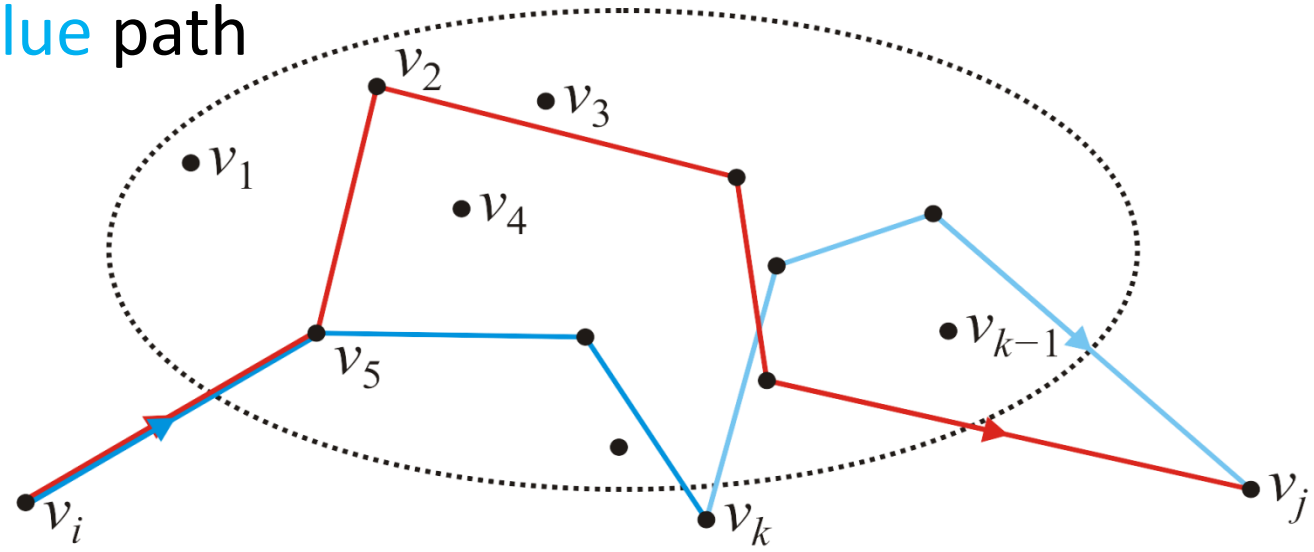
Does v_k change previous shortest path ?

With $v_1, v_2, \ldots, v_{k-1}$ as intermediates, the shortest paths from v_i to v_j, will include the paths from v_i to v_k and v_k to v_j

- The only possible shorter path including v_k would be the path from v_i to v_k *and* continuing from there to v_j

Thus, we calculate and choose: $d_{i,j}^{(k)} = \min\left\{d_{i,j}^{(k-1)}, d_{i,k}^{(k-1)} + d_{k,j}^{(k-1)}\right\}$

→Choose red or blue path



The Stages of Floyd-Warshall Algorithm

Matrix based solution :

Solve the problem stage by stage: Find

$D^{(0)}$

$D^{(1)}$

$D^{(2)}$

…

$D^{(n)}$

$D^{(k)}$ contains the shortest paths with all the intermediate vertices from the set $\{1,2\ldots,k\}$.

Why is this a DP solution ?

Each $D^{(i)}$ is stored and used in the next step.

Floyd-Warshal Algorithm :Shortest Distances

// At each step, d_{ij} is the (cost of the) shortest path
// from i to j <u>using intermediate vertices (1..k-1).</u>

$$
\begin{aligned}
&\textsc{Floyd-Warshall}(W, n)\\
&D^{(0)} \leftarrow W\\
&\textbf{for } k \leftarrow 1 \textbf{ to } n\\
&\quad \textbf{do for } i \leftarrow 1 \textbf{ to } n\\
&\qquad \textbf{do for } j \leftarrow 1 \textbf{ to } n\\
&\qquad\quad \textbf{do } d_{ij}^{(k)} \leftarrow \min\left(d_{ij}^{(k-1)}, d_{ik}^{(k-1)} + d_{kj}^{(k-1)}\right)\\
&\textbf{return } D^{(n)}
\end{aligned}
$$

- Time complexity: O(n^3) = $O(|V|^3)$
- Space complexity: O(n^3)

Floyd-Warshall Algorithm:Part of a C++ Function

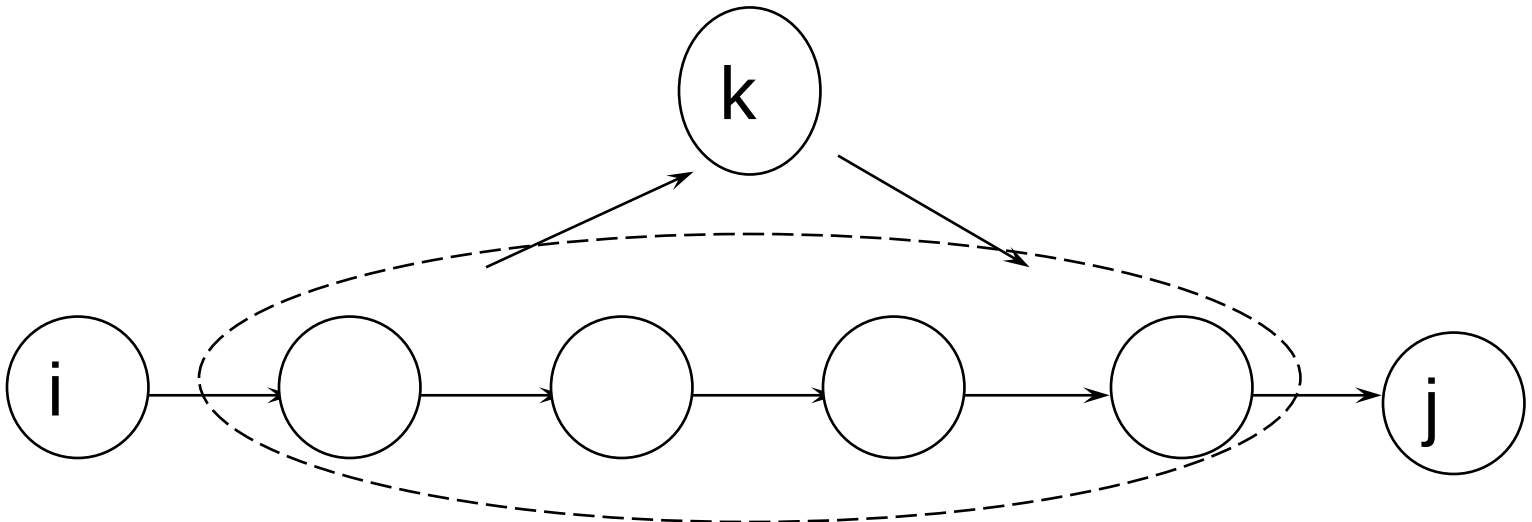
```
double d[num_vertices][num_vertices];
// Initialize the matrix
……————————————————————————————
for ( int k = 0; k < num_vertices; ++k ) {
    for ( int i = 0; i < num_vertices; ++i ) {
        for ( int j = 0; j < num_vertices; ++j ) {
            d[i][j] = min( d[i][j], d[i][k] + d[k][j] );
        }
    }
}
```

Tracing Floyd Warshall Algorithm

- Let the vertices in a graph be numbered from 1 … n.
- Consider the subset {1,2,…, k} of these n vertices.

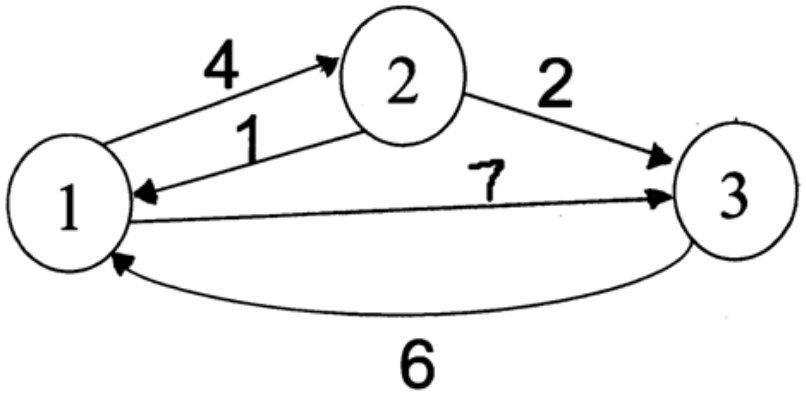
- Imagine finding the shortest path from vertex i to vertex j that uses vertices in the set {1,2,…,k} only.

- There are two situations:
  1) k is not an intermediate vertex on the shortest path.
  2) k is an intermediate vertex on the shortest path.



Tracing Floyd Warshall Algorithm: Example1

$$D^{(0)} = \begin{bmatrix} 0 & 4 & 7 \\ 1 & 0 & 2 \\ 6 & \infty & 0 \end{bmatrix}$$ =W (Original weights).



$$D^{(1)} = \begin{bmatrix} 0 & 4 & 7 \\ 1 & 0 & 2 \\ 6 & 10 & 0 \end{bmatrix}$$

k = Vertex 1:

D(3,2) = D(3,1) + D(1,2)

10= 6+4 < ∞

$$d_{ij}^{(k)} = \min(d_{ij}^{(k-1)}, d_{ik}^{(k-1)} + d_{kj}^{(k-1)})$$

$$D^{(2)} = \begin{bmatrix} 0 & 4 & 6 \\ 1 & 0 & 2 \\ 6 & 10 & 0 \end{bmatrix}$$

k=Vertex 2:

D(1,3) = D(1,2) + D(2,3)

$$D^{(3)} = \begin{bmatrix} 0 & 4 & 6 \\ 1 & 0 & 2 \\ 6 & 10 & 0 \end{bmatrix}$$

k=Vertex 3:

Nothing changes

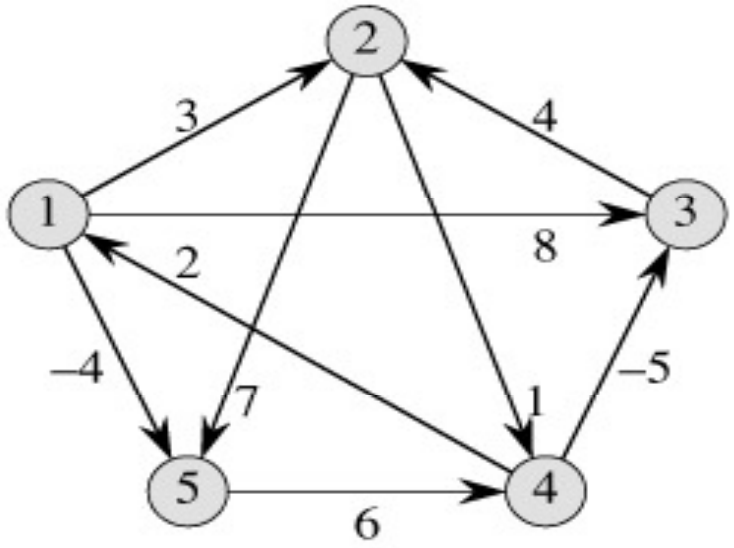
→Min distances.

Tracing Floyd Warshall Algorithm: Example2

$$d_{ij}^{(k)} = \begin{cases} w_{ij} & \text{if } k = 0 , \\ \min\left(d_{ij}^{(k-1)}, d_{ik}^{(k-1)} + d_{kj}^{(k-1)}\right) & \text{if } k \geq 1 . \end{cases}$$

k=0, the original given matrix

$$D^{(0)} = \begin{pmatrix} 0 & 3 & 8 & \infty & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & \infty & \infty \\ 2 & \infty & -5 & 0 & \infty \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix}$$



k=0

$$D^{(0)} = \begin{pmatrix} 0 & 3 & 8 & \infty & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & \infty & \infty \\ 2 & \infty & -5 & 0 & \infty \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix}$$

k=1

$$D^{(1)} = \begin{pmatrix} 0 & 3 & 8 & \infty & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & \infty & \infty \\ 2 & 5 & -5 & 0 & -2 \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix}$$

5=min(∞, 2+3)

-2=min(∞, 2+(−4))

Note that : $D^{(1)} = D^{(0)} W$

k=2

$$D^{(2)} = \begin{pmatrix} 0 & 3 & 8 & 4 & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & 5 & 11 \\ 2 & 5 & -5 & 0 & -2 \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix}$$

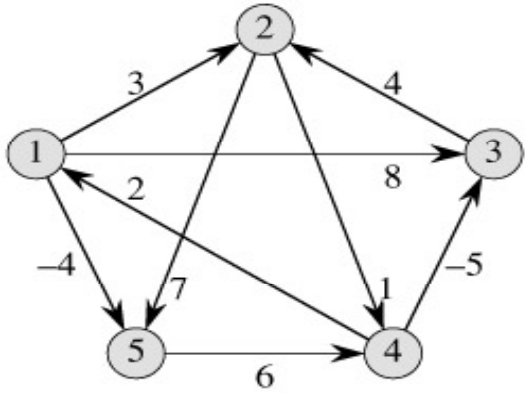
5=min(∞ ,4+1)

11=5=min(∞ ,4+7)

k=3

$$D^{(3)} = \begin{pmatrix} 0 & 3 & 8 & 4 & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & 5 & 11 \\ 2 & -1 & -5 & 0 & -2 \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix}$$

-1= min(5,-5+4)



k=4

$$D^{(4)} = \begin{pmatrix} 0 & 3 & -1 & 4 & -4 \\ 3 & 0 & -4 & 1 & -1 \\ 7 & 4 & 0 & 5 & 3 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & 5 & 1 & 6 & 0 \end{pmatrix}$$

k=5

$$D^{(5)} = \begin{pmatrix} 0 & 1 & -3 & 2 & -4 \\ 3 & 0 & -4 & 1 & -1 \\ 7 & 4 & 0 & 5 & 3 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & 5 & 1 & 6 & 0 \end{pmatrix}$$

Final shortest distances

Complexity

Time complexity, running time : 3 nested for loops, n= # of vertices, thus, $O(n^3)$.

Memory required : The same, $O(n^3)$.

If we drop the superscripts and use the same array: $O(n^2)$.

Transitive Closure

- **Transitive Closure defines a matrix which shows which pairs of vertices are reachable in a graph.**

- If a graph is given, we have to find a vertex j which is reachable from another vertex i, for all vertex pairs (i, j).

- **The final matrix is of the Boolean type. When there is a value 1 for vertex i to vertex j, it means that there is at least one path from i to j.**

- **Note that we are concerned with the existence of paths but not the path lengths.**

Transitive Closure of a Directed Graph

Which Vertices of a given directed graph are Connected?

- Computing a transitive closure of a directed graph gives complete information about which vertices are connected to which other vertices.

$$t_{ij}^{(0)} = \begin{cases} 0 & i \neq j \text{ and } (i, j) \notin E \\ 1 & i = j \text{ or } (i, j) \in E \end{cases}$$

- **Transitive closure of a directed graph G=(V,E):**

$$t_{ij}^{(k)} = t_{ij}^{(k-1)} \vee (t_{ik}^{(k-1)} \wedge t_{kj}^{(k-1)})$$

For i, j , k = 1, 2, …, n , define $t_{ij}^{(k)}=1$, if $\exists$ a path in G from i to j with all intermediate vertices in {1, 2, …, k}; otherwise $t_{ij}^{(k)}=0$

Transitive Closure of a Graph : Explanation

Input: Boolean adjacency matrix: (0-1 matrix)

→Unweighted adjacency graph *G is given*:

$W[i][j] = 1$, if $(i,j) \in E$,

$W[i][j] = 0$ otherwise.

Output: Transitive closure matrix

$T[i][j] = 1$, if there is a path of any length from i to j in G,

$T[i][j] = 0$ otherwise.

This matrix is also called the "Reachibility matrix"

- **Algorithm:**
  1. **Just run Floyd-Warshall with the adjacency matrix of G**
  2. **More efficient: use only Boolean operators**

Transitive Closure : Part of a C++ Function

The *transitive closure* is a Boolean Matrix:

```
bool tc[num_vertices][num_vertices];

// Initialize the matrix tc
//   ...

// Run Floyd-Warshall with Boolean tc matrix
   for ( int k = 0; k < num_vertices; ++k ) {
     for ( int i = 0; i < num_vertices; ++i ) {
        for ( int j = 0; j < num_vertices; ++j ) {
            tc[i][j] = tc[i][j] || (tc[i][k] && tc[k][j]);
        }
    }
}
```

Transitive Closure Algorithm: Explanation

- The algorithm is the SAME as the Floyd-Warshall Algorithm, except for when we find a non-infinity path, we simply add an edge to the transitive closure graph.
- In the algorithm , arithmetic operations are replaced by logical operations v and ^
- Reason:Logical operators are more efficient to apply.
- Every round, we use the previous route reached namely, $R^{(k-1)}$

How to generate $R^{(k)}$ from $R^{(k-1}$? There are two possibilities:

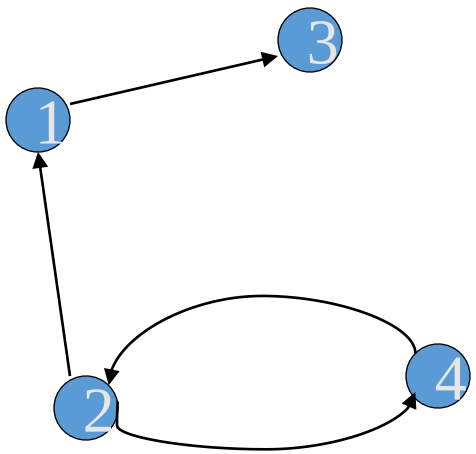
$r_{ij}^{(k)} = r_{ij}^{(k-1)}$ or (No change after adding v_k)

$= (r_{ik}^{(k-1)}$ and $r_{kj}^{(k-1)})$ (A new path is added)

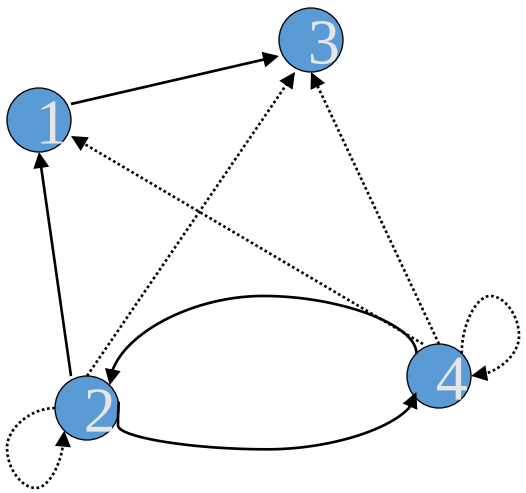
The algorithm makes this choice for every k .

Transitive Closure: Example #1

- Example Graphs and their transitive closure matrices :



$$\begin{matrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{matrix}$$



$$\begin{matrix} 0 & 0 & 1 & 0 \\ 1 & \mathbf{1} & \mathbf{1} & 1 \\ 0 & 0 & 0 & 0 \\ \mathbf{1} & 1 & \mathbf{1} & \mathbf{1} \end{matrix}$$

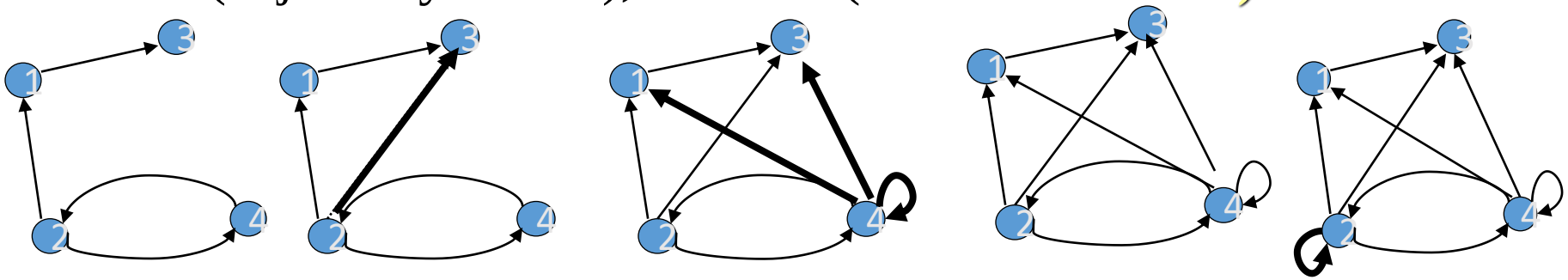
Note that there is no connection from 3 to any other node

Transitive Closure: Example #1

Constructs transitive closure T as the last matrix in the sequence of n-by-n matrices $R^{(0)}$, … , $R^{(k)}$, … , $R^{(n)}$ where

$R^{(k)}[i,j] = 1$ iff there is nontrivial path from i to j with only first k vertices allowed as intermediate

Note that $R^{(0)} = A$ (adjacency matrix), $R^{(n)} = T$ (transitive closure)



$R^{(0)}$

$$\begin{matrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{matrix}$$

$R^{(1)}$

$$\begin{matrix} 0 & 0 & 1 & 0 \\ 1 & 0 & \mathbf{1} & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{matrix}$$

$R^{(2)}$

$$\begin{matrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ \mathbf{1} & 1 & \mathbf{1} & \mathbf{1} \end{matrix}$$

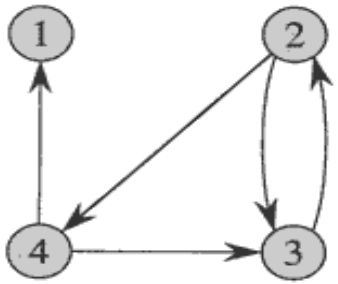
$R^{(3)}$

$$\begin{matrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{matrix}$$

$R^{(4)}$

$$\begin{matrix} 0 & 0 & 1 & 0 \\ 1 & \mathbf{1} & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{matrix}$$

Transitive Closure: Example #2



$$T^{(0)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix} \quad T^{(1)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix} \quad T^{(2)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

$$T^{(3)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \quad T^{(4)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Figure 25.5 A directed graph and the matrices $T^{(k)}$ computed by the transitive-closure algorithm.

Computer Applications of Transitive closure

- Computing the transitive closure of a digraph is an important problem in many computer science applications. Some examples:
  - Evaluation of recursive database queries.
  - Analysis of reachability (connectivity) of transition graphs in communication networks.
  - Construction of parsing automata in compilers.